

Medical Disease Symptom Reference

Type 2 Diabetes

Patients commonly experience increased thirst and frequent urination, often accompanied by unexplained weight loss despite normal or increased appetite. Fatigue, blurred vision, and slow-healing sores are frequent complaints. Many patients report tingling or numbness in the hands and feet. Recurrent infections, particularly of the skin or urinary tract, are also common indicators of poorly controlled blood glucose levels.

Hypertension

Often called the silent killer, hypertension rarely produces noticeable symptoms in its early stages. When present, symptoms may include persistent headaches, especially in the morning, along with dizziness and blurred or double vision. Some patients experience nosebleeds, shortness of breath, and chest pain. In severe cases, pounding in the chest, neck, or ears may occur, signaling dangerously elevated blood pressure levels.

Asthma

Asthma is characterized by recurrent episodes of wheezing, a whistling sound when breathing, accompanied by shortness of breath and chest tightness. Coughing, particularly at night or early morning, is a hallmark symptom. Attacks may be triggered by allergens, exercise, cold air, or respiratory infections. Between episodes, patients may feel completely normal, though airway inflammation persists and can worsen without proper management.

Chronic Obstructive Pulmonary Disease

COPD presents with a persistent productive cough, often called a smoker's cough, producing large amounts of mucus particularly in the mornings. Progressive breathlessness during daily activities is a defining symptom, worsening over time. Frequent respiratory infections and wheezing are common. Advanced stages bring cyanosis, barrel chest appearance, and significant weight loss as the body expends extra energy simply to breathe.

Rheumatoid Arthritis

Rheumatoid arthritis causes symmetrical joint pain, swelling, and stiffness, most notably in the small joints of the hands, wrists, and feet. Morning stiffness lasting more than an hour is a hallmark feature. Fatigue, low-grade fever, and loss of appetite often accompany joint symptoms. Over time, joints may become deformed, and the disease can affect other organs including the heart, lungs, and eyes in what is known as extra-articular involvement.

Osteoarthritis

Osteoarthritis produces joint pain that worsens with activity and improves with rest. Stiffness is typically brief, lasting under thirty minutes in the morning. Affected joints, commonly the knees, hips, hands, and spine, may produce a grating sensation or sound called crepitus during

movement. Swelling and tenderness around the joint are common, and bony enlargements called osteophytes may develop, causing visible joint deformity in advanced cases.

Alzheimer's Disease

Early symptoms of Alzheimer's include frequent memory lapses, particularly for recent events, and difficulty recalling names or words mid-conversation. Confusion about dates, places, or familiar faces emerges progressively. Patients may misplace objects, show poor judgment, and withdraw from social activities. As the disease advances, individuals lose the ability to perform basic daily tasks, experience significant personality changes, and may develop behavioral disturbances such as agitation.

Parkinson's Disease

The hallmark of Parkinson's disease is a resting tremor, typically beginning in one hand and described as a pill-rolling motion. Bradykinesia, or slowed movement, makes once-simple tasks time-consuming and difficult. Muscle rigidity causes stiffness and limits the range of motion. Postural instability results in a stooped posture and shuffling gait, increasing fall risk. Non-motor symptoms such as sleep disturbances, depression, and loss of smell often precede motor symptoms by years.

Multiple Sclerosis

Multiple sclerosis symptoms vary widely depending on which nerves are affected. Common presentations include fatigue, numbness or tingling in limbs, muscle weakness, and coordination problems. Vision disturbances such as blurred or double vision and optic neuritis are frequent early symptoms. Bladder dysfunction, cognitive difficulties, and heat sensitivity are also characteristic. Symptoms typically relapse and remit, though progressive forms involve steady worsening without clear recovery periods.

Epilepsy

Epilepsy is defined by recurrent unprovoked seizures, which vary widely in type and severity. Tonic-clonic seizures involve sudden loss of consciousness followed by body stiffening and rhythmic jerking. Absence seizures cause brief lapses in awareness with staring spells. Focal seizures may produce unusual sensations, involuntary movements, or altered awareness in one body region. Post-seizure confusion, fatigue, and temporary weakness on one side of the body, called Todd's paralysis, commonly follow convulsive episodes.

Migraine

Migraines typically involve severe throbbing or pulsing pain on one side of the head, often accompanied by nausea and vomiting. Heightened sensitivity to light, sound, and sometimes smell makes routine activities extremely difficult. About a third of sufferers experience aura, which may include visual disturbances like flashing lights or blind spots, tingling sensations, or speech difficulties. Attacks can last from four hours to three days and may be preceded by warning signs such as mood changes or food cravings.

Irritable Bowel Syndrome

IBS presents with chronic abdominal pain and cramping that is often relieved by defecation. Altered bowel habits, including diarrhea, constipation, or alternating between the two, are central features. Bloating and excess gas cause significant discomfort and distension. Mucus in the stool is sometimes reported. Symptoms are typically triggered or worsened by certain foods, stress, or hormonal changes, and the condition follows a relapsing and remitting course without structural damage to the intestines.

Crohn's Disease

Crohn's disease causes persistent diarrhea, often watery or bloody, accompanied by severe abdominal cramps and pain, typically in the lower right area. Unintended weight loss and malnutrition result from impaired nutrient absorption. Fever and fatigue are common during flare-ups. Perianal complications such as fistulas and abscesses occur in some patients. Extra-intestinal manifestations including joint pain, eye inflammation, and skin rashes may also be present alongside gastrointestinal symptoms.

Ulcerative Colitis

Ulcerative colitis is characterized by bloody diarrhea, often containing mucus and pus, along with urgency and inability to defecate despite the urge. Abdominal pain and cramping are concentrated in the lower left quadrant. Rectal bleeding and tenesmus, a persistent feeling of incomplete evacuation, are hallmark symptoms. During severe flares, patients may experience fever, rapid heart rate, and significant weight loss. The disease increases the long-term risk of colorectal cancer in affected individuals.

Celiac Disease

Celiac disease triggers an immune response to gluten causing diarrhea, bloating, and abdominal pain. Stools may be pale, foul-smelling, and float due to fat malabsorption. Despite eating normally, patients often experience weight loss and nutritional deficiencies leading to anemia, bone pain, and fatigue. In children, growth delays and failure to thrive are concerning signs. Dermatitis herpetiformis, an intensely itchy blistering skin rash, is an extra-intestinal manifestation unique to celiac disease.

Hypothyroidism

An underactive thyroid causes fatigue and sluggishness that is disproportionate to activity levels. Weight gain despite normal eating habits, constipation, and cold intolerance are classic symptoms. Skin becomes dry and pale, hair thins and becomes brittle, and nails grow slowly. Bradycardia and elevated cholesterol levels are common findings. In severe cases, a condition called myxedema can develop, causing facial puffiness, hoarse voice, and, if untreated, potentially life-threatening complications including coma.

Hyperthyroidism

An overactive thyroid accelerates metabolism, causing unintentional weight loss despite an increased appetite. Patients experience rapid or irregular heartbeat, nervousness, anxiety, and irritability. Heat intolerance with excessive sweating is characteristic. Tremors in the hands and fingers, increased frequency of bowel movements, and fatigue despite feeling energized are common. Graves' disease, the most common cause, can also produce eye problems including bulging eyes, double vision, and light sensitivity.

Systemic Lupus Erythematosus

Lupus is a multisystem disease whose symptoms vary greatly between individuals. The characteristic butterfly-shaped rash across the cheeks and nose is seen in many patients. Joint pain and swelling, fatigue, and fever are common. Photosensitivity causes skin rashes after sun exposure. Kidney inflammation, called lupus nephritis, can lead to protein in the urine and high blood pressure. Neurological symptoms, chest pain from pleuritis or pericarditis, and anemia further reflect the disease's wide-ranging impact.

Fibromyalgia

Fibromyalgia is defined by widespread musculoskeletal pain throughout the body, often described as a constant dull ache. Fatigue is profound and unrefreshing sleep is nearly universal. Cognitive difficulties known as fibro fog impair memory, concentration, and mental clarity. Headaches, including tension headaches and migraines, are frequent. Many patients experience irritable bowel symptoms, restless legs, and heightened sensitivity to touch, temperature, and sound, significantly impacting daily functioning and quality of life.

Gout

Gout causes sudden, severe episodes of pain, swelling, redness, and warmth in affected joints, most commonly the big toe. The skin over the joint may appear shiny and feel extremely tender to even the lightest touch. Attacks typically occur at night and can last for days to weeks. Repeated episodes can cause the deposition of urate crystals in soft tissues, forming visible lumps called tophi. Kidney stones composed of uric acid are a potential complication of chronic elevated uric acid levels.

Psoriasis

Psoriasis causes raised, inflamed patches of skin covered with silvery-white scales, most commonly appearing on the elbows, knees, scalp, and lower back. Affected areas are often itchy, sore, and may crack or bleed. Nails may become pitted, ridged, or separate from the nail bed. About a third of patients develop psoriatic arthritis, causing joint pain, stiffness, and swelling. The condition follows a cyclical pattern with flares triggered by stress, infections, certain medications, and skin injuries.

Eczema

Atopic dermatitis causes dry, intensely itchy skin that becomes red and inflamed when scratched. The itch-scratch cycle can lead to thickened, leathery patches of skin in chronic cases. In infants, rashes typically appear on the face and scalp, while in older children and adults they concentrate in the elbow and knee creases. Weeping or crusting lesions occur during flares. Patients are also prone to skin infections as the impaired skin barrier allows bacteria to penetrate, and the condition often coexists with asthma and hay fever.

Psoriatic Arthritis

Psoriatic arthritis combines the skin symptoms of psoriasis with joint inflammation. Affected joints become painful, swollen, and stiff, and the pattern can differ significantly from rheumatoid arthritis, sometimes affecting joints asymmetrically. Dactylitis, where an entire finger or toe swells like a sausage, is a distinctive feature. Enthesitis, inflammation where tendons and

ligaments attach to bone, causes heel pain and tenderness. Nail changes, including pitting and onycholysis, are common and can precede joint involvement by years.

Ankylosing Spondylitis

Ankylosing spondylitis primarily affects the spine, causing chronic pain and stiffness in the lower back and buttocks that is characteristically worse in the morning and after prolonged inactivity but improves with exercise. The pain often alternates between the two sides early in the disease. Over time, inflammation can cause vertebrae to fuse, limiting spinal mobility and producing a stooped posture. Eye inflammation called uveitis occurs in many patients. Fatigue and pain at tendon insertion sites are additional common complaints.

Systemic Sclerosis

Systemic sclerosis, or scleroderma, causes the skin to harden and tighten, initially affecting the fingers, hands, and face. Raynaud's phenomenon, where fingers and toes turn white then blue then red in response to cold or stress, is often an early sign. Difficulty swallowing and heartburn reflect esophageal involvement. Interstitial lung disease causes progressive breathlessness. Renal hypertensive crisis is a life-threatening complication. Calcinosis, telangiectasias, and joint pain round out the diverse clinical picture of this complex disease.

Sjögren's Syndrome

Sjögren's syndrome primarily affects moisture-producing glands, causing dry eyes that burn, itch, and feel gritty, as well as persistent dry mouth that makes chewing and swallowing difficult. Dental decay accelerates due to reduced saliva. Parotid gland swelling and difficulty speaking are common. Fatigue and joint pain are frequent systemic complaints. In primary Sjögren's, the disease acts independently, while secondary Sjögren's occurs alongside other autoimmune conditions. Peripheral neuropathy and kidney involvement may develop in some patients.

Vasculitis

Vasculitis encompasses inflammation of blood vessels and presents differently depending on which vessels are affected. Common symptoms include fever, fatigue, and unexplained weight loss. Skin manifestations such as purpura, livedo reticularis, and ulcers are frequent. Kidney involvement causes blood or protein in the urine and declining renal function. Lung involvement produces cough and breathlessness. Neuropathy causes tingling and weakness. Sinus congestion and nosebleeds are characteristic of granulomatosis with polyangiitis, a specific form of vasculitis.

Polymyalgia Rheumatica

Polymyalgia rheumatica causes aching and stiffness in the shoulders, neck, upper arms, hips, and thighs, which is typically bilateral and severely limits movement in the morning. The stiffness often lasts over an hour and makes activities such as raising the arms above the head or rising from a chair extremely difficult. Fatigue, mild fever, and unintentional weight loss accompany musculoskeletal symptoms. The condition occurs almost exclusively in adults over fifty and responds dramatically to corticosteroid treatment.

Giant Cell Arteritis

Giant cell arteritis causes new, severe headaches, often throbbing and located at the temples. Scalp tenderness, particularly when combing hair, and jaw pain when chewing, known as jaw claudication, are highly characteristic symptoms. Visual symptoms including sudden blurred or double vision or partial vision loss can rapidly progress to permanent blindness if untreated. Fever, fatigue, and weight loss are common systemic features. It frequently coexists with polymyalgia rheumatica and requires urgent high-dose corticosteroid treatment to prevent vision loss.

Osteoporosis

Osteoporosis itself is often asymptomatic until a fracture occurs, earning it the label of a silent disease. When fractures do occur, they most commonly affect the vertebrae, hip, and wrist. Vertebral compression fractures cause acute back pain and may lead to progressive loss of height and a stooped posture called kyphosis. Hip fractures cause severe pain, inability to bear weight, and an outwardly rotated, shortened leg. Risk factors including post-menopausal status, low bone density, and previous fractures help identify patients before fractures occur.

Pernicious Anemia

Pernicious anemia results from vitamin B12 deficiency due to lack of intrinsic factor. Fatigue and weakness are prominent early symptoms, along with shortness of breath and pallor from the resulting anemia. Neurological manifestations, including numbness and tingling in the hands and feet, difficulty walking, and balance problems, can be severe and irreversible if untreated. Glossitis, a smooth and sore tongue, and mild jaundice from red blood cell destruction may be observed. Cognitive changes and mood disturbances also occur in some patients.

Iron Deficiency Anemia

Iron deficiency anemia causes fatigue, weakness, and pale skin as a result of reduced hemoglobin levels. Shortness of breath and dizziness occur, especially during exertion. Patients may experience brittle or spoon-shaped nails, called koilonychia, and a sore, swollen tongue. Pica, the craving to eat non-food substances such as ice, clay, or starch, is a distinctive but poorly understood symptom. Cold hands and feet, frequent headaches, and a fast or irregular heartbeat are additional manifestations of significant iron deficiency.

Deep Vein Thrombosis

Deep vein thrombosis typically presents with pain, swelling, warmth, and redness in the affected limb, most commonly the calf. The pain may feel like cramping or soreness and worsens with walking or standing. A bluish discoloration of the skin may be apparent. Some cases are entirely asymptomatic and are only detected when a pulmonary embolism occurs. Risk factors include prolonged immobility, recent surgery, pregnancy, and clotting disorders. Prompt diagnosis is critical to prevent potentially fatal clot migration to the lungs.

Pulmonary Embolism

Pulmonary embolism causes sudden onset shortness of breath that can range from mild to severe. Chest pain that worsens when breathing deeply, coughing, or bending is characteristic, and patients may cough up blood-streaked sputum. Rapid heart rate and a sense of impending doom accompany severe cases. Lightheadedness and fainting can occur if blood flow to the heart is significantly impaired. Signs of deep vein thrombosis in the leg are often present

simultaneously. Massive pulmonary embolism can cause cardiovascular collapse and sudden death.

Coronary Artery Disease

Coronary artery disease classically produces chest pain or pressure, called angina, that is triggered by physical exertion or emotional stress and relieved by rest. The pain may radiate to the arm, shoulder, jaw, or back. Shortness of breath accompanies exertion and may be the sole symptom in some patients. Women and diabetic patients are more likely to experience atypical symptoms such as nausea, extreme fatigue, or back pain. Silent ischemia, with no symptoms at all, is common, and the first manifestation can tragically be a heart attack.

Heart Failure

Heart failure symptoms reflect either inadequate forward flow or backward congestion of blood. Breathlessness during activity and while lying flat, relieved by sitting upright, is highly characteristic. Ankle, foot, and leg swelling from fluid accumulation is common, as is abdominal bloating from ascites. Persistent cough or wheezing, particularly at night, may produce frothy or pink-tinged sputum. Fatigue, weakness, and reduced exercise tolerance are prominent. Rapid weight gain from fluid retention and reduced urination during the day with increased urination at night are also reported.

Atrial Fibrillation

Atrial fibrillation may cause palpitations described as a fluttering, racing, or irregular heartbeat. Many patients feel fatigued and short of breath, especially with exertion. Dizziness, lightheadedness, and chest discomfort or pressure are reported. Some patients experience no symptoms at all, with the arrhythmia discovered incidentally. The major risk associated with AF is stroke, as blood can pool in the atria and form clots. Reduced exercise tolerance and weakness reflect decreased cardiac output when the atria fail to contract efficiently.

Stroke

Stroke symptoms appear suddenly and follow the FAST acronym: facial drooping on one side, arm weakness or numbness, speech difficulties including slurred or incomprehensible words, and time to call emergency services. Additional symptoms include sudden severe headache with no known cause, sudden vision problems in one or both eyes, and dizziness or loss of balance. Depending on the brain region affected, patients may experience confusion, memory loss, or difficulty understanding language. Prompt emergency treatment is essential to minimize permanent neurological damage.

Meningitis

Meningitis classically presents with a severe headache, high fever, and neck stiffness, forming the diagnostic triad. Photophobia and phonophobia, sensitivity to light and sound, are common. Nausea, vomiting, and confusion or altered consciousness reflect severe illness. Bacterial meningitis may cause a distinctive non-blanching purpuric rash indicating septicemia, which is a medical emergency. In infants, symptoms may be subtler, including bulging fontanelle, high-pitched crying, poor feeding, and mottled skin. Rapid antibiotic treatment is critical to survival.

Pneumonia

Pneumonia presents with cough productive of thick, discolored sputum, high fever with shaking chills, and pleuritic chest pain that worsens with deep breathing or coughing. Shortness of breath and rapid breathing reflect the degree of lung involvement. Fatigue, loss of appetite, and confusion, particularly in elderly patients, are common systemic features. Physical examination reveals reduced breath sounds and crackles. Severity varies from patients who can be managed at home to those requiring intensive care with mechanical ventilation.

Tuberculosis

Pulmonary tuberculosis causes a persistent cough lasting three or more weeks, which may produce blood-stained sputum. Night sweats that are drenching and unexplained weight loss are classic systemic symptoms. Low-grade fever and fatigue are chronic companions. Chest pain and breathlessness develop as the disease progresses. In extrapulmonary tuberculosis, symptoms depend on the site affected, including lymph node swelling, back pain in spinal TB, and severe headache and neurological signs in tuberculous meningitis.

HIV/AIDS

Early HIV infection may cause an acute flu-like illness with fever, swollen lymph nodes, sore throat, rash, muscle aches, and mouth sores. After this initial phase, patients may be asymptomatic for years. As immunity declines, opportunistic infections emerge, including recurrent thrush, pneumocystis pneumonia, and toxoplasmosis. Unexplained weight loss exceeding ten percent of body weight, chronic diarrhea, and persistent fever are AIDS-defining conditions. Night sweats, fatigue, and neurological complications including dementia are also seen in advanced disease.

Malaria

Malaria classically causes cyclical episodes of fever, chills, and sweating that recur every two to three days depending on the parasite species. These attacks are often preceded by a cold stage with shivering, followed by a hot stage with high fever and headache, then a sweating stage as the temperature falls. Nausea, vomiting, muscle pain, and fatigue accompany each episode. Severe malaria, most commonly caused by *Plasmodium falciparum*, can cause cerebral involvement, severe anemia, kidney failure, respiratory distress, and rapid death if untreated.

Dengue Fever

Dengue fever causes sudden high fever, severe headache, and pain behind the eyes that worsens with eye movement. Intense muscle and joint pain give it the nickname breakbone fever. A rash appears two to five days after fever onset. Mild bleeding manifestations such as nosebleeds, gum bleeding, and easy bruising may occur. Dengue hemorrhagic fever is a severe form causing plasma leakage, severe bleeding, and organ failure. Thrombocytopenia, leukopenia, and elevated liver enzymes are characteristic laboratory findings.

Typhoid Fever

Typhoid fever begins with gradually rising fever over several days, reaching up to forty degrees Celsius. Headache, malaise, loss of appetite, and abdominal pain are early features. A relative bradycardia, where heart rate fails to rise proportionally with fever, is characteristic. Rose spots, faint salmon-colored spots on the trunk, appear in some patients. Diarrhea or constipation may

occur. Without treatment, serious complications including intestinal perforation and hemorrhage, septicemia, and encephalitis can develop in the third and fourth weeks.

Cholera

Cholera produces sudden onset profuse, watery diarrhea described as rice-water stools because of their pale, watery appearance containing mucus flakes. This is followed by projectile vomiting. Rapid, severe dehydration occurs, causing intense thirst, muscle cramps particularly in the legs, reduced urination, and sunken eyes. Skin loses its elasticity in a sign called poor skin turgor. In severe cases, the voice becomes hoarse, skin turns blue-grey, and blood pressure drops critically. Without fluid replacement, death can occur within hours of symptom onset.

Hepatitis A

Hepatitis A initially causes flu-like symptoms including fatigue, nausea, vomiting, abdominal pain, and low-grade fever. Dark urine and pale stools appear before visible jaundice develops as liver inflammation impairs bilirubin processing. Jaundice, a yellow discoloration of the skin and whites of the eyes, is accompanied by intense itching. Loss of appetite and a general feeling of illness precede and accompany jaundice. Most patients recover fully within two months, though a small number develop fulminant hepatic failure requiring liver transplantation.

Hepatitis B

Chronic hepatitis B may be silent for decades while liver damage accumulates. When symptomatic, patients experience fatigue, loss of appetite, jaundice, dark urine, and abdominal pain in the upper right quadrant from liver enlargement. Acute infection adds fever, joint pain, and a skin rash. Advanced disease with cirrhosis causes fluid accumulation in the abdomen, leg swelling, easy bruising, and mental confusion from hepatic encephalopathy. Liver cancer risk is substantially elevated in chronic carriers, particularly those with ongoing viral replication.

Hepatitis C

Hepatitis C is often called a silent infection because most people have minimal or no symptoms during the acute phase. When symptoms do occur they include fatigue, fever, nausea, and abdominal discomfort. Chronic infection progresses slowly over years, causing liver fibrosis and eventually cirrhosis. Patients with cirrhosis develop fatigue, itching, jaundice, portal hypertension leading to varices and ascites, and elevated risk of liver cancer. Extra-hepatic manifestations include cryoglobulinemia causing skin rash, neuropathy, and kidney disease.

Cirrhosis

Cirrhosis symptoms in early stages may be minimal, with fatigue and weakness being the most common complaints. As the disease progresses, patients develop jaundice, itching, and easy bruising from impaired clotting factor synthesis. Portal hypertension causes varices, which can bleed catastrophically, and ascites, a fluid-filled abdomen. Spider angiomas and palmar erythema are cutaneous signs. Hepatic encephalopathy produces confusion, sleep disturbances, and ultimately coma. Male patients may develop gynecomastia and testicular atrophy from hormonal disruption.

Pancreatitis

Acute pancreatitis causes severe, persistent pain in the upper abdomen that often radiates straight through to the back. The pain typically worsens after eating, especially fatty foods, and may be partially relieved by leaning forward. Nausea and vomiting are common. Fever and rapid pulse reflect systemic inflammation. In severe cases, bruising around the navel or flanks signals internal bleeding. Chronic pancreatitis adds steatorrhea, oily foul-smelling stools from enzyme deficiency, and eventually diabetes as the endocrine pancreas becomes damaged.

Appendicitis

Appendicitis classically begins with diffuse abdominal pain around the navel that migrates over hours to the lower right quadrant, where it becomes localized and severe. Nausea and vomiting follow the onset of pain. Low-grade fever and loss of appetite are characteristic. Rebound tenderness, where pain worsens upon sudden release of applied abdominal pressure, and guarding of the abdominal muscles are physical signs. Diarrhea or constipation may occur. A perforated appendix causes sudden brief pain relief followed by worsening pain and signs of peritonitis.

Gallstones

Gallstones may be asymptomatic or cause biliary colic, a severe, steady pain in the upper right abdomen or center of the abdomen that may radiate to the right shoulder or back. Episodes typically occur after fatty meals and last from thirty minutes to several hours before resolving. Nausea and vomiting accompany the pain. Fever and jaundice indicate complications such as cholecystitis or common bile duct obstruction. Gallstone-related pancreatitis causes pain radiating to the back and elevated pancreatic enzyme levels.

Kidney Stones

Kidney stones cause sudden, severe, cramping pain in the back and side below the ribs, radiating to the lower abdomen and groin as the stone moves. The pain fluctuates in waves as the ureter contracts around the stone. Nausea and vomiting accompany severe pain. Hematuria, blood in the urine, causes pink, red, or brown discoloration. Frequent urination, urgency, and burning during urination suggest the stone is approaching the bladder. Fever and chills indicate a urinary tract infection, which requires urgent treatment to prevent urosepsis.

Urinary Tract Infection

Urinary tract infections cause a burning or stinging sensation during urination, along with increased frequency and urgency. The urine may appear cloudy, dark, or blood-tinged and often has a strong, unpleasant odor. Pelvic pain or pressure in the lower abdomen is common in bladder infections. Upper tract involvement, or pyelonephritis, adds fever, chills, back and flank pain, nausea, and vomiting. In elderly patients, UTIs may present atypically with confusion, falls, or general deterioration rather than classic urinary symptoms.

Prostate Cancer

Early prostate cancer is often asymptomatic and detected through PSA screening. As it advances, urinary symptoms emerge including increased frequency, urgency, weak or interrupted urine stream, difficulty starting urination, and incomplete bladder emptying. Blood in the urine or semen may occur. Bone metastases cause deep, persistent pain, most commonly in the back, hips, and pelvis. Unexplained weight loss and fatigue reflect advanced disease.

Spinal cord compression from metastatic spread can cause neurological symptoms including leg weakness and bowel or bladder dysfunction.

Breast Cancer

Breast cancer may first manifest as a painless lump or thickening in the breast or underarm. Changes in breast size or shape, skin dimpling or puckering resembling orange peel, and nipple retraction or discharge other than breast milk are warning signs. Redness, scaling, or flaking of the nipple or breast skin warrants evaluation. In inflammatory breast cancer, the breast becomes red, warm, and swollen rapidly. Pain is not a reliable early symptom, but persistent breast pain in a specific location may occasionally indicate malignancy.

Lung Cancer

Lung cancer symptoms include a new, persistent cough or a change in a chronic cough. Coughing up blood or rust-colored sputum is an alarming sign requiring immediate evaluation. Chest pain that worsens with deep breathing, coughing, or laughing may be present. Shortness of breath and wheezing reflect airway involvement or pleural effusion. Hoarseness from recurrent laryngeal nerve compression is characteristic of left-sided tumors. Unexplained weight loss, loss of appetite, and fatigue are systemic symptoms. Superior vena cava syndrome causes facial and arm swelling.

Colorectal Cancer

Colorectal cancer may present with a change in bowel habits, including persistent diarrhea, constipation, or a feeling of incomplete evacuation. Rectal bleeding or blood in the stool, which may be bright red or dark, requires investigation. Abdominal discomfort, cramping, or pain that does not resolve is concerning. Narrow or ribbon-like stools may indicate a tumor obstructing the colon. Unexplained weight loss and fatigue from chronic blood loss causing anemia are systemic signs. In advanced disease, an abdominal mass may be palpable.

Ovarian Cancer

Ovarian cancer is often called a silent killer because early symptoms are vague and easily attributed to other causes. Bloating, pelvic or abdominal pain, difficulty eating or feeling full quickly, and urinary urgency or frequency are the most common symptoms. These may be persistent and represent a change from usual patterns. Fatigue, indigestion, back pain, and abnormal menstrual cycles may also occur. Unfortunately, most cases are diagnosed at an advanced stage because the ovaries' deep pelvic location makes early detection difficult without specific screening.

Cervical Cancer

Early cervical cancer may produce abnormal vaginal bleeding, particularly after sexual intercourse, between periods, or after menopause. An unusual vaginal discharge that may be watery, bloody, and have a foul odor is another warning sign. As the tumor grows, pelvic pain, pain during intercourse, and lower back pain may develop. Advanced disease causes leg swelling from lymph node obstruction, difficulty urinating, and rectal bleeding from invasion of adjacent structures. Regular Pap smears detect precancerous changes before symptoms develop, enabling prevention.

Leukemia

Leukemia symptoms reflect the failure of normal blood cell production. Fatigue, weakness, and pallor result from anemia. Easy bruising, prolonged bleeding from cuts, and petechiae, tiny red or purple spots under the skin, indicate thrombocytopenia. Recurrent or severe infections signal impaired immunity. Bone and joint pain from leukemic infiltration of the marrow is common. Lymph node enlargement, spleen and liver enlargement causing abdominal fullness, weight loss, and night sweats are additional features. Fever without an identifiable infection is characteristic.

Lymphoma

Lymphoma typically presents with painless swelling of lymph nodes in the neck, armpits, or groin. B symptoms, which include unexplained fever, drenching night sweats, and unintentional weight loss of more than ten percent of body weight over six months, indicate a more aggressive disease course. Persistent fatigue, itchy skin, and decreased appetite are common. In mediastinal lymphoma, chest pain, cough, and superior vena cava syndrome may develop. Abdominal lymphoma causes fullness, pain, or altered bowel habits. Splenomegaly contributes to early satiety.

Multiple Myeloma

Multiple myeloma causes bone pain, most commonly in the back, hips, and skull, from osteolytic lesions that weaken the skeleton. Pathological fractures, particularly vertebral compression fractures, cause severe pain and height loss. Fatigue and weakness stem from anemia. Recurrent bacterial infections, especially pneumonia, result from suppressed normal immunoglobulin production. Kidney dysfunction from light chain deposition causes protein in the urine. Hypercalcemia produces nausea, thirst, constipation, and confusion. Amyloidosis can develop as a complication.

Pancreatic Cancer

Pancreatic cancer in the head of the pancreas commonly causes painless jaundice with dark urine and pale stools as bile flow is obstructed. Abdominal pain that radiates to the back, new-onset or worsening diabetes, and significant weight loss are concerning features. Nausea, loss of appetite, and digestive problems from exocrine insufficiency occur. Migratory thrombophlebitis, recurrent blood clots in different veins, may be an early paraneoplastic sign. The prognosis is poor due to late presentation, lack of early symptoms, and resistance to most treatments.

Stomach Cancer

Gastric cancer in its early stages often produces only vague symptoms such as indigestion, heartburn, and mild abdominal discomfort that are easily attributed to benign conditions. As the tumor grows, patients experience persistent abdominal pain, loss of appetite, and progressive weight loss. Nausea and vomiting, sometimes with blood in the vomit, are concerning features. Difficulty swallowing occurs with tumors at the gastroesophageal junction. Black or tarry stools and fatigue indicate occult gastrointestinal bleeding causing iron deficiency anemia.

Esophageal Cancer

Esophageal cancer classically presents with progressive dysphagia, beginning with difficulty swallowing solid foods and advancing to problems with liquids. This progression over weeks to months often prompts medical evaluation. Unintentional weight loss is invariably significant by the time of diagnosis. Odynophagia, painful swallowing, may accompany dysphagia. Regurgitation of undigested food, chronic cough, hoarseness from recurrent laryngeal nerve involvement, and hiccups may occur. Vomiting blood signals bleeding from a tumor ulceration or involvement of major blood vessels.

Bladder Cancer

Bladder cancer most commonly presents with painless hematuria, blood in the urine that may cause visible discoloration or be detected only microscopically. Urinary symptoms such as increased frequency, urgency, and burning during urination mimic a urinary tract infection. Pelvic pain develops as the tumor invades surrounding structures. Advanced disease causes flank pain from ureteral obstruction, bone pain from metastases, and unintentional weight loss. Repeated urinary tract infections that do not respond to treatment should prompt investigation to exclude underlying bladder pathology.

Thyroid Cancer

Thyroid cancer often presents as a painless lump or nodule in the neck that the patient or their doctor notices on routine examination. The thyroid gland may feel enlarged. Difficulty swallowing or a sensation of something in the throat may occur if the nodule compresses the esophagus. Hoarseness or voice changes result from recurrent laryngeal nerve involvement. Neck lymph node enlargement suggests regional spread. Most thyroid cancers grow slowly and are highly treatable, but aggressive subtypes such as anaplastic thyroid cancer progress rapidly with poor prognosis.

Skin Cancer (Melanoma)

Melanoma presents as changes in an existing mole or as a new pigmented lesion. The ABCDE criteria guide evaluation: Asymmetry, where one half does not match the other; Border irregularity with ragged or notched edges; Color variation including shades of brown, black, red, or white; Diameter greater than six millimeters; and Evolution, meaning any change in size, shape, or color. Melanoma may also appear as a dark streak under a nail. Metastatic melanoma causes systemic symptoms including unexplained weight loss, fatigue, and symptoms related to the specific organs involved.

Endometrial Cancer

Endometrial cancer most commonly presents with abnormal uterine bleeding, which in postmenopausal women means any vaginal bleeding and in premenopausal women means unusually heavy, prolonged, or between-period bleeding. Watery or blood-tinged vaginal discharge may occur. Pelvic pain or pressure, difficulty urinating, and pain during intercourse develop as the tumor grows. The high rate of early symptomatic presentation, particularly postmenopausal bleeding, typically leads to diagnosis at an earlier, more treatable stage compared to other gynecological cancers with less distinctive early symptoms.

Kidney Cancer

Kidney cancer may remain asymptomatic for a long time, often detected incidentally on imaging performed for other reasons. When symptoms do appear, the classic triad consists of flank pain, visible blood in the urine, and a palpable abdominal mass, though this triad together is now uncommon at diagnosis. Persistent pain in the side or back that does not go away, unexplained weight loss, and fatigue are common presentations. Fever without an obvious infection, elevated blood pressure, and anemia or polycythemia may occur as paraneoplastic phenomena.

Liver Cancer

Primary liver cancer symptoms are often masked by underlying cirrhosis. Abdominal pain or tenderness in the right upper quadrant, unexplained weight loss, and loss of appetite are common. Abdominal swelling from tumor growth or worsening ascites may occur. New jaundice in a cirrhotic patient or sudden deterioration in liver function should prompt investigation. An abdominal bruit, a vascular sound heard through a stethoscope, may be audible over the tumor. Fever, nausea, and early satiety from an enlarged liver are additional symptoms. Paraneoplastic hypoglycemia can cause episodes of low blood sugar.

Testicular Cancer

Testicular cancer typically presents as a painless lump or swelling in one testicle, often noticed by the patient. A feeling of heaviness or dragging in the scrotum may be described. Occasional dull aching or discomfort in the affected testicle or lower abdomen can occur, though sharp pain is unusual unless there is bleeding into the tumor. Gynecomastia, breast tissue enlargement in men, results from hormone secretion by certain tumor types. Back pain and shortness of breath may indicate retroperitoneal or pulmonary metastases respectively in advanced disease.

Pheochromocytoma

Pheochromocytoma causes episodes of hypertension, headache, sweating, and palpitations caused by excess catecholamine release from the adrenal gland tumor. These classic paroxysms may be triggered by physical activity, emotional stress, or certain medications. Between episodes, blood pressure may be persistently elevated or normal. Pallor, anxiety, tremor, and a feeling of impending doom accompany attacks. Weight loss, heat intolerance, and elevated blood sugar may suggest the diagnosis. A sustained or paroxysmal hypertensive crisis with the above symptoms should prompt hormonal investigation.

Addison's Disease

Addison's disease develops insidiously with fatigue, muscle weakness, and loss of appetite being the earliest and most common symptoms. Weight loss, nausea, diarrhea, and abdominal pain follow. A distinctive feature is hyperpigmentation, a darkening of the skin and mucous membranes particularly in areas exposed to friction or sun. Dizziness upon standing due to low blood pressure and salt cravings from sodium wasting are characteristic. An Addisonian crisis is life-threatening, causing severe vomiting, diarrhea, abdominal pain, extreme weakness, low blood pressure, and shock.

Cushing's Syndrome

Cushing's syndrome from excess cortisol causes central obesity with a rounded moon face, fatty hump between the shoulders called a buffalo hump, and large purplish stretch marks on the

abdomen, thighs, and arms. Muscle weakness, particularly proximal muscle wasting, causes difficulty rising from a chair. Thin, fragile skin bruises easily and heals slowly. Hypertension, elevated blood sugar, and osteoporosis leading to fractures are metabolic complications. Women may experience irregular menstrual periods and excess facial hair. Depression, anxiety, and cognitive impairment are psychiatric manifestations.

Acromegaly

Acromegaly from excess growth hormone causes gradual enlargement of the hands, feet, and facial features. Rings and shoes may no longer fit, and patients may notice their face has changed significantly. The jaw and brow protrude, the nose widens, and gaps develop between the teeth. Joint pain and carpal tunnel syndrome from tissue swelling are common. Excessive sweating, oily skin, and deepening of the voice result from soft tissue changes. Sleep apnea develops from airway soft tissue enlargement. Headaches, visual field defects, and amenorrhea reflect pituitary tumor mass effect.

Hypopituitarism

Hypopituitarism produces symptoms corresponding to the deficient hormones. Growth hormone deficiency causes fatigue, reduced muscle mass, increased body fat, and impaired quality of life. Thyroid-stimulating hormone deficiency produces hypothyroid symptoms. Gonadotropin deficiency causes loss of libido, erectile dysfunction in men, menstrual irregularity or cessation in women, and infertility. Corticotropin deficiency leads to symptoms similar to Addison's disease but without hyperpigmentation. Prolactin deficiency prevents lactation postpartum and is often the earliest sign of pituitary damage.

Polycystic Ovary Syndrome

Polycystic ovary syndrome presents with menstrual irregularities including infrequent, prolonged, or absent periods reflecting anovulation. Excess androgen causes hirsutism, male-pattern hair growth on the face, chest, and back, along with acne and scalp hair thinning. Polycystic ovaries may cause pelvic discomfort. Difficulty losing weight and central obesity are common, as are signs of insulin resistance. Infertility results from irregular or absent ovulation. Long-term risks include type 2 diabetes, cardiovascular disease, endometrial cancer, and metabolic syndrome.

Endometriosis

Endometriosis causes chronic pelvic pain that worsens during menstruation, called dysmenorrhea, and may be debilitating. Deep pain during or after sexual intercourse, called dyspareunia, is characteristic. Painful bowel movements or urination during menstrual periods suggest deep infiltrating disease. Heavy or irregular bleeding, including spotting between periods, is common. Fatigue during menstrual periods can be severe. Infertility affects many patients. The severity of symptoms does not correlate with the extent of disease, and diagnosis is often significantly delayed due to symptom normalization.

Premenstrual Syndrome

Premenstrual syndrome causes a predictable cluster of physical and emotional symptoms occurring in the days before menstruation and resolving shortly after it begins. Physical symptoms include bloating, breast tenderness, headaches, fatigue, and food cravings.

Emotional symptoms such as mood swings, irritability, anxiety, and depression can be significant. Sleep disturbances and difficulty concentrating add to functional impairment. In severe cases, classified as premenstrual dysphoric disorder, symptoms are so intense that they substantially disrupt daily life, work, and relationships.

Menopause

Menopause, defined as twelve months after the final menstrual period, is preceded by a transition called perimenopause with irregular periods. Hot flashes, a sudden feeling of warmth spreading over the upper body accompanied by flushing and sweating, are the most common symptom. Night sweats disrupt sleep, contributing to fatigue. Vaginal dryness causes discomfort during intercourse. Urinary symptoms such as urgency and recurrent infections reflect urogenital atrophy. Mood changes, difficulty concentrating, and joint pain are common. Long-term estrogen decline increases risk of osteoporosis and cardiovascular disease.

Benign Prostatic Hyperplasia

Benign prostatic hyperplasia causes lower urinary tract symptoms as the enlarged prostate obstructs the urethra. Obstructive symptoms include weak urine stream, difficulty starting urination, intermittent flow, straining, and a sensation of incomplete bladder emptying. Irritative symptoms include urinary frequency, urgency, nocturia, and urge incontinence. Urinary retention, the inability to void, is an acute complication requiring catheterization. Recurrent urinary tract infections, bladder stones, and upper tract damage can develop in neglected or severe cases over time.

Erectile Dysfunction

Erectile dysfunction is characterized by persistent difficulty achieving or maintaining an erection sufficient for satisfactory sexual activity. It can range from occasional difficulty to complete inability to achieve an erection. Associated symptoms may include reduced sexual desire, premature ejaculation, and anxiety around sexual performance. Underlying conditions such as cardiovascular disease, diabetes, hypertension, and depression frequently contribute. Nocturnal penile tumescence, which is normal erection during sleep, helps distinguish psychogenic from organic causes. The condition significantly impacts self-esteem, relationships, and quality of life.

Infertility

Infertility, defined as failure to achieve pregnancy after twelve months of unprotected intercourse, may have no direct symptoms itself. However, underlying causes produce their own symptoms. Women with ovulatory disorders experience irregular or absent menstrual cycles. Endometriosis and pelvic inflammatory disease cause pelvic pain. Uterine fibroids may cause heavy periods. In men, hormonal imbalances, varicoceles, or previous infections may be asymptomatic or cause testicular discomfort and altered sexual function. Hormonal imbalances in either partner may cause reduced libido and other endocrine symptoms.

Anxiety Disorder

Generalized anxiety disorder causes excessive, uncontrollable worry about everyday matters that is difficult to control. Physical symptoms include restlessness, feeling on edge, muscle tension, headaches, and fatigue. Concentration difficulties and irritability are common cognitive and emotional features. Sleep disturbances, particularly difficulty falling asleep and staying

asleep due to racing thoughts, significantly impair functioning. In panic disorder, discrete episodes of intense fear accompanied by chest pain, shortness of breath, dizziness, tingling, and a sense of losing control or dying occur suddenly and unexpectedly.

Depression

Major depressive disorder is characterized by a persistently depressed mood or pervasive loss of interest and pleasure in activities once enjoyed, present for most of the day, nearly every day for at least two weeks. Changes in appetite and weight, sleep disturbances either excessive or insufficient, and psychomotor agitation or retardation are common. Fatigue and loss of energy are prominent. Difficulty concentrating, making decisions, and feelings of worthlessness or excessive guilt impair daily functioning. In severe cases, recurrent thoughts of death or suicidal ideation may be present.

Bipolar Disorder

Bipolar disorder involves episodes of mania or hypomania alternating with episodes of depression. Manic episodes are characterized by elevated or irritable mood, markedly decreased need for sleep, racing thoughts, pressured speech, increased goal-directed activity, and impulsive risky behaviors such as spending sprees or sexual indiscretions. Grandiosity and inflated self-esteem are prominent. Depressive episodes mirror major depressive disorder with profound low mood, fatigue, hopelessness, and suicidal thoughts. Mixed episodes with simultaneous manic and depressive features can be particularly severe and dangerous.

Schizophrenia

Schizophrenia is characterized by positive symptoms including hallucinations, most commonly auditory voices commenting on the person's behavior or conversing with each other, and delusions, which are fixed false beliefs such as being persecuted or having special powers. Disorganized thinking produces incoherent speech and behavior. Negative symptoms include emotional blunting, poverty of speech, social withdrawal, and diminished motivation. Cognitive symptoms impair attention, memory, and executive functioning. The illness typically begins in late adolescence or early adulthood and follows a chronic, often relapsing course.

Obsessive-Compulsive Disorder

Obsessive-compulsive disorder involves intrusive, unwanted obsessive thoughts, images, or urges that cause significant anxiety and distress. Common obsessions include fears of contamination, harming others, or making mistakes. To neutralize the anxiety, patients perform compulsive behaviors, which may include repetitive handwashing, checking, counting, or arranging objects. These rituals are time-consuming, interfere with daily activities, and provide only temporary relief. Patients recognize their obsessions as irrational but are unable to resist them. Avoidance of triggering situations further constricts functioning.

Post-Traumatic Stress Disorder

PTSD develops after exposure to a traumatic event and is characterized by intrusive re-experiencing of the trauma through flashbacks and nightmares that feel vivid and terrifying. Persistent avoidance of trauma-related stimuli, including people, places, and memories, leads to emotional numbing and social withdrawal. Hyperarousal manifests as an exaggerated startle response, irritability, anger outbursts, difficulty concentrating, and sleep disturbances. Negative

changes in thoughts and mood, including persistent negative beliefs, distorted blame, and emotional detachment, are additional diagnostic criteria for this debilitating condition.

Attention Deficit Hyperactivity Disorder

ADHD presents with inattention, hyperactivity, and impulsivity that are inappropriate for developmental level and interfere with functioning across multiple settings. Inattentive symptoms include difficulty sustaining attention, following instructions, and completing tasks, along with frequent loss of belongings and forgetfulness. Hyperactive symptoms manifest as fidgeting, inability to remain seated, excessive talking, and difficulty engaging in quiet activities. Impulsivity causes interrupting others, difficulty awaiting turns, and hasty decision-making. In adults, hyperactivity may be less obvious but inattention and emotional dysregulation persist.

Autism Spectrum Disorder

Autism spectrum disorder is characterized by persistent difficulties in social communication and interaction across multiple contexts. Deficits include reduced reciprocal social-emotional engagement, difficulty with nonverbal communication such as eye contact and facial expressions, and challenges developing and maintaining relationships. Restricted, repetitive behaviors include insistence on sameness, ritualized routines, stereotyped movements, fixated interests, and unusual sensory responses such as hypersensitivity to sounds, textures, or lights. Symptoms are present from early childhood, though may not fully manifest until social demands exceed capacities.

Eating Disorder (Anorexia Nervosa)

Anorexia nervosa is characterized by extreme restriction of caloric intake leading to significantly low body weight, intense fear of gaining weight, and a distorted perception of one's own body shape. Physical consequences include lanugo, fine downy body hair; dry skin; brittle nails; hair loss; amenorrhea; cold intolerance; and severe fatigue. Bradycardia and hypotension reflect cardiac compromise. Electrolyte imbalances, particularly hypokalemia, cause muscle weakness and dangerous arrhythmias. Osteoporosis develops from hormonal disruption and nutritional deficiency. Cognitive impairment and depressive symptoms are common comorbidities.

Insomnia

Insomnia is characterized by difficulty falling asleep, staying asleep, or waking too early and being unable to return to sleep, occurring at least three nights per week for at least three months. Daytime consequences include fatigue, low energy, and impaired concentration and memory, which significantly affect occupational and social functioning. Mood disturbances including irritability, depression, and anxiety are common. Behavioral changes such as increased reliance on caffeine, daytime napping, and spending excessive time in bed paradoxically perpetuate the insomnia cycle. Chronic insomnia increases risk for several medical and psychiatric conditions.

Sleep Apnea

Obstructive sleep apnea is characterized by loud, disruptive snoring interspersed with episodes of complete silence during apneas, followed by gasping or choking sounds as breathing resumes. Bed partners often notice these episodes before the patient is aware. Excessive daytime sleepiness despite adequate sleep time is the most common daytime symptom.

Morning headaches, dry mouth, and memory and concentration difficulties reflect fragmented, non-restorative sleep. Nocturia, frequent nighttime urination, is common. Long-term consequences include hypertension, atrial fibrillation, metabolic syndrome, and increased cardiovascular mortality.

Restless Legs Syndrome

Restless legs syndrome causes an overwhelming, unpleasant urge to move the legs, often accompanied by uncomfortable sensations described as crawling, creeping, pulling, aching, or tingling deep within the limbs. Symptoms are triggered or worsened by rest and inactivity, particularly in the evening and at night. Movement, particularly walking, provides temporary relief. The condition severely disrupts sleep, causing insomnia and excessive daytime fatigue. Periodic limb movements during sleep, involuntary leg jerking, are frequently associated. Quality of life is substantially impaired, and mood disorders are a common comorbidity.

Narcolepsy

Narcolepsy is characterized by excessive daytime sleepiness with sudden, irresistible sleep attacks that occur inappropriately during eating, conversation, or driving. Cataplexy, a sudden loss of muscle tone triggered by strong emotions such as laughter or surprise, causes the person to collapse while fully conscious. Sleep paralysis, a temporary inability to move upon waking or falling asleep, and hypnagogic hallucinations, vivid dream-like experiences when drowsy, complete the classic tetrad. Disrupted nighttime sleep despite daytime somnolence is paradoxical. The condition is caused by loss of hypocretin-producing neurons.

Chronic Kidney Disease

Chronic kidney disease is often asymptomatic in early stages, with symptoms emerging as kidney function progressively declines. Fatigue, weakness, and decreased mental sharpness result from anemia and uremia. Edema in the legs, ankles, and feet from reduced sodium and water excretion becomes apparent. Decreased urine output, foamy urine from proteinuria, and blood in the urine may occur. Persistent nausea, vomiting, and loss of appetite characterize uremia. Itching from mineral deposits in the skin, muscle cramps, shortness of breath from fluid in the lungs, and hypertension are additional manifestations.

Acute Kidney Injury

Acute kidney injury presents with decreased urine output, though output may sometimes be preserved. Fluid retention causes swelling of the legs, ankles, and face. Shortness of breath from pulmonary edema may develop rapidly. Fatigue, confusion, and drowsiness reflect uremia and metabolic disturbances. Nausea, vomiting, and loss of appetite are common. In severe cases, dangerous electrolyte imbalances including hyperkalemia cause cardiac arrhythmias. Chest pain may occur from pericarditis in uremia. The cause determines additional specific symptoms such as rash in allergic interstitial nephritis or oliguria following shock.

Nephrotic Syndrome

Nephrotic syndrome is defined by massive proteinuria, leading to severe hypoalbuminemia and subsequent edema, particularly marked periorbital puffiness upon waking and dependent leg edema worsening throughout the day. Frothy or foamy urine from high protein content is a characteristic sign. Hyperlipidemia and lipiduria develop as compensatory liver responses.

Anasarca, generalized body swelling including ascites and pleural effusions, occurs in severe cases. Increased infection susceptibility, particularly encapsulated bacteria, and hypercoagulability causing venous thromboembolism are significant complications of severe nephrotic syndrome.

Lupus Nephritis

Lupus nephritis, kidney inflammation from systemic lupus, may be clinically silent initially but detected by urinalysis showing blood and protein. As it progresses, hypertension develops and requires management. Edema of the ankles and around the eyes reflects protein loss and fluid retention. Frothy urine indicates significant proteinuria. Declining kidney function causes fatigue, nausea, and eventually uremic symptoms. Worsening lupus nephritis may be signaled by a flare of systemic lupus symptoms including rash, joint pain, and fever. Regular monitoring of urinalysis and kidney function is essential in all lupus patients.

Polycystic Kidney Disease

Polycystic kidney disease causes progressive bilateral renal enlargement from multiple cysts, often producing a sense of fullness, pain, or discomfort in the flanks or abdomen. Hematuria from cyst rupture or bleeding is common. Hypertension develops early, often before kidney function declines, from activation of the renin-angiotensin system. Urinary tract infections and kidney stones are frequent complications. Progressive renal impairment leads to chronic kidney disease symptoms. Intracranial aneurysms affect a significant proportion of patients, and cysts may also develop in the liver, pancreas, and seminal vesicles.

Glomerulonephritis

Glomerulonephritis presents with hematuria, causing cola or tea-colored urine from red blood cell casts, and proteinuria detected on urinalysis. Hypertension and edema develop as sodium and water retention occur. Acute nephritic syndrome following a streptococcal infection causes sudden hematuria, hypertension, and oliguria. Rapidly progressive glomerulonephritis causes swift decline in kidney function over days to weeks with significant constitutional symptoms. Depending on the underlying cause, patients may also have systemic features such as rash, joint pain, or respiratory symptoms.

Sickle Cell Disease

Sickle cell disease causes recurrent painful vaso-occlusive crises, with severe pain in the bones, chest, and abdomen resulting from sickled red blood cells obstructing small blood vessels. Acute chest syndrome presents with chest pain, fever, and respiratory distress and is a leading cause of death. Chronic anemia causes fatigue, pallor, and jaundice. Stroke, including silent cerebral infarction, occurs in children. Splenic sequestration causes sudden, severe anemia. Avascular necrosis of the femoral head causes progressive hip pain and disability. Leg ulcers and priapism are additional complications.

Hemophilia

Hemophilia causes prolonged and excessive bleeding after injuries or surgeries, spontaneous bleeding into joints called hemarthrosis causing pain, swelling, and warmth, and spontaneous muscle hematomas. Repeated joint bleeds lead to chronic arthropathy with joint damage, pain, and limited mobility. Easy bruising and prolonged bleeding from minor cuts are characteristic.

Intracranial hemorrhage, though less common, is potentially fatal. Severe hemophilia produces symptoms from early infancy when circumcision or heel stick sampling reveals prolonged bleeding. The severity of symptoms correlates with clotting factor activity levels.

Von Willebrand Disease

Von Willebrand disease is the most common inherited bleeding disorder, causing mucosal and skin bleeding rather than the deep tissue bleeding of hemophilia. Prolonged nosebleeds lasting more than ten minutes, easy bruising, heavy and prolonged menstrual periods, and prolonged bleeding after dental procedures or minor cuts are characteristic. Postpartum hemorrhage is a concern in affected women. Gastrointestinal bleeding may occur. Surgical and traumatic bleeding can be severe. Most cases are mild and may go undiagnosed until a hemostatic challenge reveals the abnormality through excessive bleeding.

Thalassemia

Beta-thalassemia major presents in infancy with severe hemolytic anemia causing pallor, jaundice, and profound fatigue. Growth retardation and failure to thrive occur without treatment. Extramedullary hematopoiesis causes hepatosplenomegaly and characteristic facial bone deformities including frontal bossing and prominent cheekbones from medullary expansion. Iron overload from hemolysis and transfusions causes cardiac failure, liver cirrhosis, and endocrine dysfunction. Without regular transfusions and chelation therapy, survival beyond the second decade was historically rare for this severe form.

Kawasaki Disease

Kawasaki disease, primarily affecting young children, begins with a high fever persisting for five or more days that does not respond to antibiotics. The characteristic features include red eyes without discharge, red and cracked lips, a strawberry tongue, diffuse redness of the mouth and throat, a polymorphous rash over the trunk, redness and swelling of the hands and feet followed by skin peeling, and swollen lymph nodes in the neck. The most serious complication is coronary artery aneurysm formation, which can cause myocardial infarction in children. Irritability is particularly prominent.

Cystic Fibrosis

Cystic fibrosis produces thick, sticky mucus that obstructs the lungs and digestive tract. Persistent cough producing thick sputum, frequent lung infections including pneumonia and bronchiectasis, and progressive decline in lung function are the major respiratory manifestations. Wheezing and exercise intolerance develop. Pancreatic exocrine insufficiency causes malabsorption, leading to bulky, greasy stools, poor weight gain despite good appetite, and nutritional deficiencies. Meconium ileus in newborns, male infertility from absent vas deferens, and sinus disease are additional features of this systemic disease.

Huntington's Disease

Huntington's disease causes a triad of motor, cognitive, and psychiatric symptoms that progressively worsen. Motor symptoms begin with subtle involuntary movements called chorea, which are random, uncontrollable jerking movements, along with impaired coordination and balance. Writing, speaking, and swallowing become progressively difficult. Cognitive decline encompasses impaired executive function, slowed processing, and dementia. Psychiatric

symptoms including depression, anxiety, irritability, and obsessive-compulsive behaviors often precede or accompany motor symptoms. The disease is uniformly fatal, typically within ten to twenty years of symptom onset.

Amyotrophic Lateral Sclerosis

ALS causes progressive muscle weakness and wasting from simultaneous upper and lower motor neuron degeneration. Early symptoms often include hand weakness making fine motor tasks difficult, foot drop causing tripping, or slurred speech. Muscle fasciculations, visible twitching under the skin, and muscle cramps are common early features. As the disease progresses, weakness spreads to involve limb, bulbar, and respiratory muscles. Difficulty swallowing and choking become life-threatening. Most patients retain cognitive function, though a minority develop frontotemporal dementia. Respiratory failure is the eventual cause of death.

Myasthenia Gravis

Myasthenia gravis causes fluctuating muscle weakness that characteristically worsens with activity and improves with rest. Ocular symptoms, including ptosis, drooping of one or both eyelids, and diplopia, double vision, are the most common presenting features. Bulbar weakness causes difficulty chewing, swallowing, and speaking, with a nasal quality to the voice. Facial weakness gives a flat expression. Limb weakness affects proximal muscles causing difficulty climbing stairs. Myasthenic crisis, sudden severe respiratory muscle weakness, is life-threatening and requires immediate mechanical ventilation.

Guillain-Barré Syndrome

Guillain-Barré syndrome presents with progressive ascending weakness beginning in the legs and moving upward, often preceded by a respiratory or gastrointestinal infection. Tingling and numbness in the hands and feet typically precede or accompany weakness. The weakness can progress rapidly over days to weeks, potentially involving respiratory muscles and causing life-threatening respiratory failure. Deep tendon reflexes are absent or diminished. Autonomic instability causes fluctuating blood pressure, heart rate abnormalities, and urinary retention. Most patients recover substantially, though the process is slow and incomplete recovery occurs in some.

Polymyositis

Polymyositis causes progressive, symmetrical proximal muscle weakness affecting the shoulders, upper arms, hips, and thighs, making activities like rising from a chair, climbing stairs, and lifting objects above the head increasingly difficult. Dysphagia and dysphonia occur with pharyngeal muscle involvement. Muscle pain and tenderness are present in some patients. Interstitial lung disease causes progressive breathlessness. Cardiac involvement includes myocarditis and arrhythmias. Elevated muscle enzymes, particularly creatine kinase, confirm muscle inflammation. Constitutional symptoms of fever, fatigue, and weight loss accompany the myopathy.

Dermatomyositis

Dermatomyositis combines the proximal muscle weakness of polymyositis with characteristic skin manifestations. The heliotrope rash, a purple-red discoloration of the eyelids often with periorbital edema, is highly specific. Gottron's papules, raised violet scaly patches over the

knuckles, elbows, and knees, are pathognomonic. A V-sign rash on the anterior chest and shawl sign on the upper back and shoulders reflect photosensitive distribution. Mechanic's hands, with rough, cracked skin on the fingers, suggest antisynthetase syndrome. Malignancy is associated with adult-onset dermatomyositis in a significant proportion of cases.

Marfan Syndrome

Marfan syndrome affects connective tissue throughout the body. Skeletal manifestations include tall stature with disproportionately long limbs, fingers, and toes; scoliosis; pectus deformities; and joint hypermobility with frequent dislocations. Ocular features include lens dislocation causing visual impairment and myopia. Cardiovascular abnormalities are the most life-threatening, with aortic root dilatation progressing to aortic aneurysm and dissection causing sudden severe tearing chest and back pain. Mitral valve prolapse may cause palpitations. Spontaneous pneumothorax and dural ectasia causing back pain are additional manifestations.

Ehlers-Danlos Syndrome

Ehlers-Danlos syndrome encompasses a group of connective tissue disorders with joint hypermobility causing frequent sprains, dislocations, and chronic joint pain as the most universal features. Skin is hyperextensible, velvety, and fragile, healing poorly and leaving wide atrophic scars after minor injuries. Easy bruising is common. Chronic musculoskeletal pain is often severe and debilitating. Autonomic dysfunction causes orthostatic intolerance and fatigue. In the vascular type, spontaneous rupture of medium and large arteries, the uterus, or bowel are life-threatening complications. Gastrointestinal dysmotility causes chronic digestive symptoms.

Wilson's Disease

Wilson's disease, a copper metabolism disorder, causes hepatic, neurological, and psychiatric manifestations. Liver disease ranges from elevated liver enzymes to cirrhosis and liver failure with jaundice, ascites, and coagulopathy. Neurological symptoms include dysarthria, drooling, difficulty swallowing, movement disorders including tremor, chorea, and dystonia, and gait abnormalities. Psychiatric symptoms such as personality changes, depression, anxiety, and psychosis may precede neurological symptoms. Kayser-Fleischer rings, golden-brown rings at the corneal periphery visible on slit-lamp examination, are a hallmark diagnostic finding.

Hemochromatosis

Hereditary hemochromatosis causes iron accumulation in multiple organs producing varied symptoms. Fatigue and joint pain, particularly affecting the second and third metacarpophalangeal joints of the hands, are common early complaints. Liver involvement causes hepatomegaly, elevated enzymes, and eventually cirrhosis with its complications. Diabetes mellitus from pancreatic islet cell damage causes typical hyperglycemic symptoms. Cardiomyopathy causes heart failure and arrhythmias. Bronze skin pigmentation from iron and melanin deposition gives the classic appearance. Hypogonadism causes loss of libido and amenorrhea in women and impotence in men.

Galactosemia

Classic galactosemia in newborns presents shortly after starting breast or formula feeding with poor feeding, vomiting, and jaundice that worsens rapidly. Lethargy and hypotonia develop, and prolonged neonatal jaundice is a warning sign. Escherichia coli sepsis is a life-threatening early

complication related to immunological dysfunction. Cataracts may develop in the first weeks of life from galactitol accumulation in the lens. Without dietary galactose restriction, liver failure with cirrhosis, intellectual disability, and neurological damage progress. Even with treatment, long-term complications include speech difficulties, cognitive deficits, and ovarian failure in females.

Phenylketonuria

Untreated phenylketonuria causes progressive intellectual disability, behavioral problems, and seizures from phenylalanine accumulation in the brain. Affected infants appear normal at birth but develop symptoms if dietary treatment is not initiated, typically through newborn screening. Hyperactivity, tremors, and autistic behaviors emerge in childhood. A characteristic mousy or musty body odor from phenylacetic acid excretion is noted. Fair skin and hair compared to unaffected family members result from reduced melanin synthesis. Maternal PKU syndrome causes birth defects in offspring of women with poorly controlled disease during pregnancy.

Neurofibromatosis Type 1

Neurofibromatosis type 1 is diagnosed by the presence of six or more café-au-lait spots, flat pale brown skin patches, along with other features. Neurofibromas, benign nerve tumors, appear as soft flesh-colored bumps under the skin and can occur internally. Lisch nodules, benign iris hamartomas visible on slit-lamp examination, cause no visual impairment but aid diagnosis. Optic pathway gliomas may impair vision. Learning disabilities and attention difficulties are common. Scoliosis, tibial dysplasia, and short stature are skeletal features. Malignant peripheral nerve sheath tumors are a serious complication in some patients.

Tuberous Sclerosis

Tuberous sclerosis causes benign tumor-like growths called hamartomas in multiple organs. Skin manifestations include hypopigmented ash-leaf spots visible under ultraviolet light, angiofibromas producing a butterfly rash on the face, shagreen patches on the lower back, and periungual fibromas around the nails. Brain tubers cause epilepsy, which is often severe and treatment-resistant, and intellectual disability. Subependymal giant cell astrocytomas can obstruct cerebrospinal fluid flow. Cardiac rhabdomyomas are common in fetuses and infants. Renal angiomyolipomas can bleed. Pulmonary lymphangiomyomatosis affects young women.

Down Syndrome

Down syndrome, caused by trisomy 21, presents at birth with characteristic features including upward-slanting eyes with epicanthal folds, a flat facial profile, small ears, a protruding tongue due to hypotonia and a relatively small mouth, a single palmar crease, and short stature. Hypotonia is universal in infancy. Intellectual disability ranges from mild to moderate. Congenital heart defects, particularly atrioventricular septal defects, occur in nearly half of all affected individuals. Hypothyroidism, hearing loss, vision problems, and atlantoaxial instability are additional medical concerns throughout life.

Turner Syndrome

Turner syndrome in girls is characterized by short stature and primary ovarian insufficiency leading to delayed or absent puberty, amenorrhea, and infertility. Lymphedema of the hands

and feet at birth, webbed neck, low posterior hairline, widely spaced nipples, and a broad chest are classical dysmorphic features. Coarctation of the aorta and bicuspid aortic valve are important cardiac abnormalities requiring monitoring. Hearing loss, recurrent middle ear infections, and learning difficulties in spatial and mathematical reasoning are common. Mosaic forms have milder presentations. Hormone replacement is necessary for feminization and bone health.

Klinefelter Syndrome

Klinefelter syndrome in males is often not diagnosed until adolescence or adulthood when infertility or delayed puberty prompts evaluation. Boys may be tall with disproportionately long legs and have learning difficulties and speech delays in childhood. Adolescent features include small testes and penis, reduced or absent puberty, gynecomastia, and sparse body and facial hair. Adults experience infertility from azoospermia, reduced libido, and erectile dysfunction from testosterone deficiency. Fatigue, osteoporosis, and metabolic syndrome are associated. Increased risk of autoimmune diseases and certain malignancies such as breast cancer is recognized.